



WACE Examination, Sample 2008

ENGINEERING STUDIES

Section 2, Electrical, electronic systems

Stage 2

Provisional Marking Guidelines

Section 2: Electrical, electronic systems

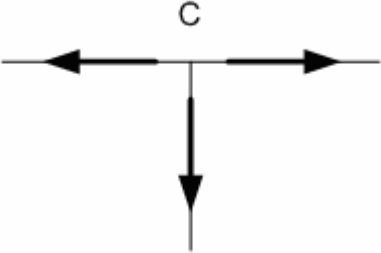
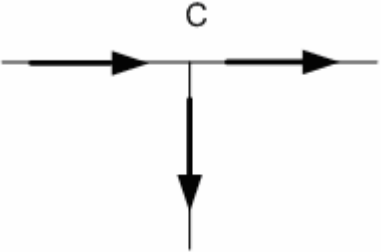
Part A: Multiple Choice Questions

1	A	potential difference across a power supply or component	1
2	B	5V	1
3	D	adjust the voltage across the components	1
4	A	create an open circuit if the current is too large	1
5	C	LDR	1
6	D	see (D) diagram on exam paper	1
7	B	protects components from current flowing in the wrong direction	1
8	C	12V	1
9	D	42 μF	1
10	A	3.43 Ω	1
			TOTAL 10

Part B: Written

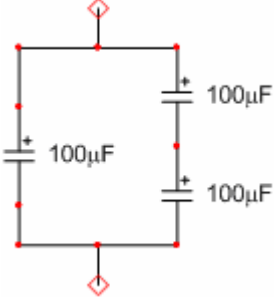
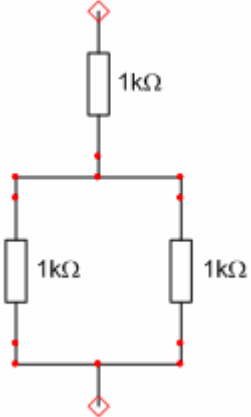
Section 2 — Electrical, electronic systems		
1a	$R_1 = 40 + 20$ $= 60\Omega$ $R_2 = R_1 \parallel R$ Therefore: $\frac{1}{R_2} = \frac{1}{R_1} + \frac{1}{R}$ $= \frac{1}{60} + \frac{1}{120}$ $= \frac{3}{120} = \frac{1}{40}$ Therefore: $R_2 = 40\Omega$ Finally: $R_T = 10 + R_2$ $= 10 + 40 = 50\Omega$	1 1 1 1 1 1 1 1 TOTAL 8
1b	(i) $V_D = 20 \times I$ $\therefore I = \frac{V_D}{20}$ $\therefore I = \frac{2}{20} = 0.1A$	1 1 TOTAL 2

	(ii) $P_{20} = \frac{V_D^2}{20}$ $= \frac{2^2}{20} = 0.2 \text{ W}$	1 1 TOTAL 2
	(iii) $V_C = 40 \times I + 20 \times I$ $= 60 \times 0.1$ $= 6 \text{ V}$	1 1 TOTAL 2
	(iv) $I_{120} = \frac{6}{120} = 0.05 \text{ A}$ Now $I_1 = I_{120} + I$ $= 0.05 + 0.1$ $= 0.15 \text{ A}$	1 1 1 TOTAL 3
	(v) $V_S = 10 \times I_1 + V_C$ $= 10 \times 0.15 + 6.0$ $= 7.5 \text{ V}$	1 1 1 TOTAL 3
1c	(i) If the 120Ω resistor is replaced by a short-circuit then the node voltage $V_C = 0$. This also means that V_D will be zero: $V_D = 0$	2 2 TOTAL 4
1c	(ii) If the 120Ω resistor is replaced by an open-circuit, then the circuit simplifies to a simple loop. Now the resistor-divider rule can be used to find V_D : $V_D = \frac{20}{20 + 40 + 10} \times 24 = 6.857 \text{ V}$ NOTE: For ease of explanation, the simple loop shown below may be used to explain the effect of the open circuit.	2 2 TOTAL 4

1d	The sum of the currents leaving node C equals zero: the current flowing to the left through the 10Ω resistor, plus the current flowing to the right through the 20Ω resistor, plus the current flowing down through R, equals zero.  Alternatively, the current that enters from the left branch is equal to the sum of the currents that leave the right and lower branches: 	1 1 TOTAL 2
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Section 2 — Electrical, electronic systems		
2a	(i) $\text{Efficiency} = \frac{\text{Power out}}{\text{Power in}} \times 100$ $\therefore 80 = \frac{100kW}{P_{in}} \times 100$ $\therefore P_{in} = \frac{100kW \times 100}{80} = 125 kW$ Energy (in joules) = Power (watts) x time (seconds) Therefore: $E = P_{in} \times t$ $= 125000 \times (15 \times 60)$ $= 125000 \times 900$ $= 112.5 MJ$	2 1 1 1 1 1 TOTAL 7
2a	(ii) Energy (in kWh) = Power (kW) x time (hours) Therefore:	1

	$E = P_{in} \times t$ $= 125 \times 7$ $= 875 \text{ kWh}$	1 1 TOTAL 3
2b	(i) To provide a current path from the positive terminal of the AC supply to the positive terminal of the DC supply, for when the AC supply is positive, then diode D2 must be reversed, as shown below. To provide a current path from the negative terminal of the AC supply to the positive terminal of the DC supply, for when the AC supply is negative (and so its negative terminal becomes more positive than its positive terminal), then diode D4 must be reversed, and shown below.	2 2 TOTAL 4
	(ii) $I_{RDC} = \frac{12}{1500} = 0.008 \text{ A}$	1 TOTAL 1
	(iii) No DC current passes through the capacitor, so $I_{CDC} = 0 \text{ A}$	1 TOTAL 1
	(iv) $P_R = \frac{12^2}{1500} = 0.096 \text{ W}$ or $P_R = I_{RDC}^2 \times 1500 = 0.008^2 \times 1500 = 0.096 \text{ W}$ or $P_R = 12 \times I_{RDC} = 12 \times 0.008 = 0.096 \text{ W}$	1 TOTAL 1
	(v) No power is dissipated in either the ideal diodes or the capacitor, so the power supplied by the AC supply must equal the power delivered to the resistor: $P_{AC} = P_R = 0.096 \text{ W}$	1 TOTAL 1
	(vi) Capacitors add in parallel. Now we can write: $150\mu\text{F} = 100\mu\text{F} + 50\mu\text{F}$ The $50\mu\text{F}$ capacitor can be obtained from the $100\mu\text{F}$ capacitors by placing two capacitors in series: $\frac{1}{C_s} = \frac{1}{100\mu\text{F}} + \frac{1}{100\mu\text{F}} = \frac{1}{50\mu\text{F}}$ The final combination is as shown below:	1 1

		2 TOTAL 4
	(vii) Resistors add in series. Now we can write: $1500\Omega = 1000\Omega + 500\Omega$ The 500Ω resistor can be obtained from the 1000Ω resistors by placing two resistors in parallel: $\frac{1}{R_p} = \frac{1}{1000\Omega} + \frac{1}{1000\Omega} = \frac{1}{500\Omega}$ The final combination is as shown below: 	1 1 2 TOTAL 4
2c	A signal from an electronic/electrical circuit (usually a low voltage circuit) causes a current to pass through the coil. This current induces a magnetic field about the soft iron core, pulling the armature down. This moves the common from the NC contact to the NO contact. This will either switch on or off an external circuit. When the coil is deactivated, the spring on the armature will pull the common back to the normally closed contract.	1 1 1 1 TOTAL 4

Section 2 — Electrical, electronic systems		
3a	With the switch closed, the switch acts as a short-circuit, and so the voltage drop across the $10k\Omega$ resistor is 12V:	1
	$I_R = \frac{12 - 0}{10000} = 1.2 \times 10^{-3} A$	2

	With short-circuit across the base-emitter terminals of the transistor, the transistor will be off, so that $I_B = 0 A$ From Kirchhoff's current law: $I_R = I_B + I_{SW}$ $\therefore I_{SW} = I_R - I_B = 1.2 \times 10^{-3} A$ As the transistor is OFF, the diode current must be zero: $I_D = 0 A$	1 1 1 1 2 1 TOTAL 10
3b	With the switch open: $I_{SW} = 0 A$ With the switch open the transistor turns on and the voltage drop across the base-emitter terminals equals 0.7V. The current through the 10kΩ resistor is now given by: $12 = 10000 \times I_R + 0.7$ $\therefore I_R = \frac{12 - 0.7}{10000} = 1.13 \times 10^{-3} A$ From Kirchhoff's current law: $I_R = I_B + I_{SW}$ $\therefore I_B = I_R - I_{SW} = 1.13 \times 10^{-3} A$ The transistor will be saturated, so that the voltage drop across collector-emitter terminals will be 0V. The current I_D can be found from Kirchhoff's voltage law: $12 = V_{LED} + 200 \times I_D + 0$ $\therefore I_D = \frac{12 - V_{LED}}{200} = \frac{12 - 2.0}{200} = 0.05 A$	1 1 2 1 1 1 1 2 1 TOTAL 11
3c	The simplest solution is to place the LED in parallel with the transistor, as shown below. LED connected from C to ground 200Ω resistor connected between +12V supply and C When the switch is closed the transistor is OFF, and all of I_{200} will flow through the LED. When the switch is open the transistor will be saturated and the voltage drop across the collector-emitter terminals will drop to zero. As this is also the voltage drop across the LED, then the LED must be turned off.	2 2

